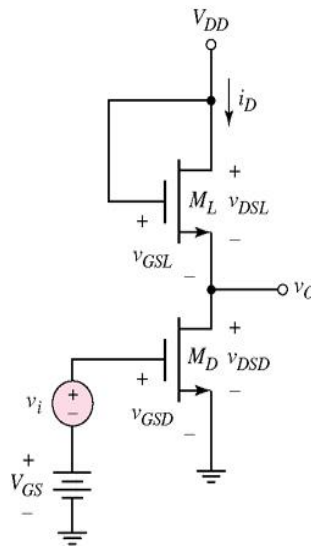


2023-2024
2nd year /Electronic I
Electronic and Telecommunication Department.
Engineering Science/ Kufa University

Single Stage Integrated MOSFET Amplifiers

NMOS Amplifiers with Enhancement Load

Example 1: Find the $(W/L)_D$ of NMOS amplifier with an enhancement load as in Figure below with a small signal voltage gain of $|A_v| = 10$. The circuit is biased at $V_{DD} = 5\text{ V}$, the transistors parameters are $V_{TN} = 1\text{ V}$, $k'_{nD} = k'_{nL}$, $\lambda = 0$ and $(W/L)_L = 1$.



Solution:

$$\approx -\frac{g_{mD}}{g_{mL}} = -\sqrt{\frac{K_{nD}}{K_{nL}}} = -\sqrt{\frac{(W/L)_D}{(W/L)_L}}$$

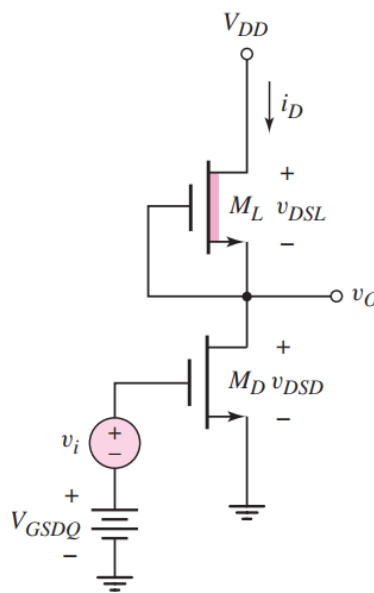
$$|A_v| = 10 = \sqrt{\frac{(W/L)_D}{(W/L)_L}}$$

$$\left(\frac{W}{L}\right)_D = 100 \left(\frac{W}{L}\right)_L$$

Then $(W/L)_D = 100$

NMOS amplifier with depletion load

Example 2: Determine the small signal voltage gain of an NMOS amplifier with depletion load for the circuit shown in the figure below, assume the transistor parameters $V_{TND} = +0.8 \text{ V}$, $V_{TNL} = -1.5 \text{ V}$, $K_{nD} = 1 \text{ mA/V}^2$, $K_{nL} = 0.2 \text{ mA/V}^2$, and $\lambda_D = \lambda_L = 0.01 \text{ V}^{-1}$ with $I_{DQ} = 0.2 \text{ mA}$.



Solution:

$$g_{mD} = 2\sqrt{K_{nD}I_{DQ}} = 2\sqrt{(1)(0.2)} = 0.894 \text{ mA/V},$$

Since $\lambda_D = \lambda_L$,

$$r_{oD} = r_{oL} = \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.01)(0.2)} = 500 \text{ k}\Omega$$

$$A_v = -g_{mD}(r_{oD} \parallel r_{oL}) = -(0.894)(500 \parallel 500) = -224$$

NMOS amplifier with active loads

CMOS common source amplifier

Example 3:

Determine the small signal voltage gain of the CMOS amplifier for the circuit shown in the figure below. Assume transistor parameters $V_{TN} = +0.8 \text{ V}$, $V_{TP} = -0.8 \text{ V}$, $k'_n = 80 \mu\text{A/V}^2$, $k'_p = 40 \mu\text{A/V}^2$, $(W/L)_n = 15$, $(W/L)_p = 30$, and $\lambda_D = \lambda_L = 0.01 \text{ V}^{-1}$. Also, assume $I_{Bias} = 0.2 \text{ mA}$.

Solution:

$$g_{mn} = 2\sqrt{K_n I_{DQ}} = 2\sqrt{\left(\frac{k'_n}{2}\right) \left(\frac{W}{L}\right)_n I_{Bias}}$$

$$= 2\sqrt{\left(\frac{0.08}{2}\right) (15)(0.2)} = 0.693 \text{ mA/V}$$

Since $\lambda_n = \lambda_p$ then the output resistances are

$$r_{on} = r_{op} = \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.01)(0.2)} = 500 \text{ k}\Omega$$

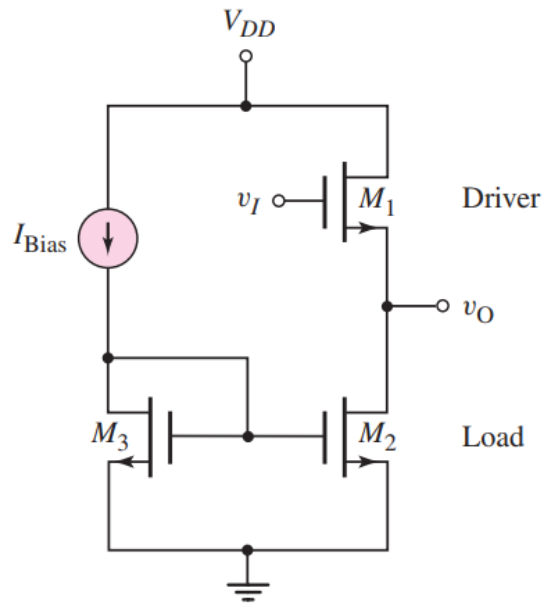
The small signal voltage gain is then obtained

$$A_v = -g_m(r_{on} \parallel r_{op}) = -(0.693)(500 \parallel 500) = -173$$

CMOS Source-Follower Amplifier

Example 4: Determine the small signal voltage gain and output impedance of the source amplifier shown in the figure below. Assume the reference bias current is

$I_{\text{Bias}} = 0.20 \text{ mA}$ and the bias voltage is $V_{\text{DD}} = 3.3 \text{ V}$. Assume that all transistors are identical with the parameters $V_{\text{TN}} = 0.4 \text{ V}$, $K_n = 0.20 \text{ mA/V}^2$, and $\lambda = 0.01 \text{ V}^{-1}$.



Solution:

$$A_v = \frac{V_o}{V_i} = \frac{g_{m1}(r_{o1} \parallel r_{o2})}{1 + g_{m1}(r_{o1} \parallel r_{o2})}$$

$$g_{m1} = 2\sqrt{K_n I_{D1}} = 2\sqrt{(0.20)(0.20)} = 0.40 \text{ mA/V}$$

$$r_{o1} = r_{o2} = \frac{1}{\lambda I_D} = \frac{1}{(0.01)(0.20)} = 500 \text{ k}\Omega$$

$$A_v = \frac{(0.40)(500 \parallel 500)}{1 + (0.40)(500 \parallel 500)}$$

$$A_v = 0.990$$

$$R_o = \frac{1}{0.40} \parallel 500 \parallel 500$$

$$R_o = 2.48 \text{ k}\Omega$$

CMOS common gate amplifier

Example 5: Determine the small-signal voltage gain and output resistance of the common-gate circuit shown in Figure below.

Assume the reference bias current is $I_{\text{Bias}} = 0.20 \text{ mA}$ and the bias voltage is $V_{\text{DD}} = 3.3 \text{ V}$. Assume that the transistor parameters are $V_{\text{TN}} = +0.4 \text{ V}$, $V_{\text{TP}} = -0.4 \text{ V}$, $K_n = 0.20 \text{ mA/V}^2$, $K_p = 0.20 \text{ mA/V}^2$, and $\lambda_n = \lambda_p = 0.01 \text{ V}^{-1}$.

M2 and M3 are matched transistors and have the same source-to-gate voltage, the bias current in M1 is $I_{\text{D1}} = I_{\text{Bias}} = 0.20 \text{ mA}$.

Solution:

$$A_v = \frac{\left(g_{m1} + \frac{1}{r_{o1}}\right)}{\left(\frac{1}{r_{o2}} + \frac{1}{r_{o1}}\right)}$$

$$g_{m1} = 2\sqrt{K_n I_{\text{D1}}} = 2\sqrt{(0.20)(0.20)} = 0.40 \text{ mA/V}$$

$$r_{o1} = r_{o2} = \frac{1}{\lambda I_{\text{D1}}} = \frac{1}{(0.01)(0.20)} = 500 \text{ k}\Omega$$

$$A_v = \frac{\left(0.40 + \frac{1}{500}\right)}{\left(\frac{1}{500} + \frac{1}{500}\right)}$$

$$A_v = 101$$

$$R_o = \frac{V_x}{I_x} = r_{o1} \parallel r_{o2} = 500 \parallel 500$$

$$R_o = 250 \text{ k}$$

Multistage Amplifier: Cascade Circuit:

Example 6: Determine the small signal voltage gain of a multi-stage cascade circuit for the circuit below. The transistor parameters are $K_{n1} = 0.5 \text{ mA/V}^2$, $K_{n2} =$

0.2 mA/V^2 and $V_{TN1} = V_{TN2} = 1.2 \text{ V}$. The quiescent drain currents are $I_{D1} = 0.2 \text{ mA}$ and $I_{D2} = 0.5 \text{ mA}$.

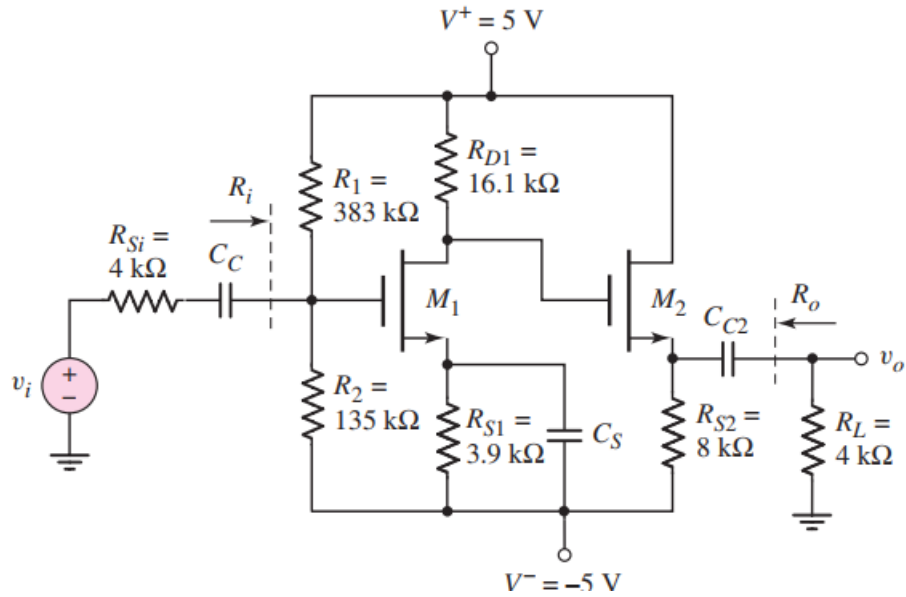
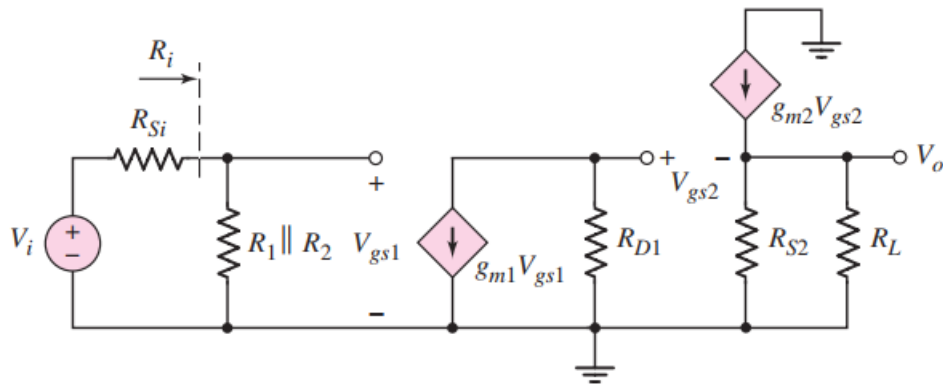


Figure 4.9 Common-source amplifier in cascade with source follower

Small-signal equivalent circuit of NMOS cascade circuit



$$g_{m1} = 2\sqrt{K_{n1}I_{D1}} = 2\sqrt{(0.5)(0.2)} = 0.632 \text{ mA/V}$$

$$g_{m2} = 2\sqrt{K_{n2}I_{D2}} = 2\sqrt{(0.2)(0.5)} = 0.632 \text{ mA/V}$$

$$V_o = g_{m2}V_{gs2}(R_{S2} \parallel R_L)$$

$$V_{gs2} + V_o = -g_{m1}V_{gs1} R_{D1}$$

$$V_{gs1} = \left(\frac{R_i}{R_i + R_{Si}} \right) \cdot V_i$$

$$V_{gs2} = -g_{m1} R_{D1} \left(\frac{R_i}{R_i + R_{Si}} \right) \cdot V_i - V_o$$

$$V_o = g_{m2} \left[-g_{m1} R_{D1} \left(\frac{R_i}{R_i + R_{Si}} \right) \cdot V_i - V_o \right] (R_{S2} \parallel R_L)$$

The small-signal voltage gain is then

$$A_v = \frac{V_o}{V_i} = \frac{-g_{m1} g_{m2} R_{D1} (R_{S2} \parallel R_L)}{1 + g_{m2} (R_{S2} \parallel R_L)} \cdot \left(\frac{R_i}{R_i + R_{Si}} \right)$$

$$A_v = \frac{-(0.632)(0.632)(16.1)(8 \parallel 4)}{1 + (0.632)(8 \parallel 4)} \cdot \left(\frac{100}{100 + 4} \right) = -6.14$$

Multistage Amplifier: Cascode Circuit

Example 7: Determine the small signal voltage gain of the cascode circuit as in the figure below. Transistor parameters are $K_{n1} = K_{n2} = 0.8 \text{ mA/V}^2$ and $V_{TN1} = V_{TN2} = 1.2 \text{ V}$. The drain current is $I_D = 0.4 \text{ mA}$ in each transistor.

Solution: As both transistors with the same I_D and K_n then

$$g_{m1} = g_{m2} = 2\sqrt{K_n I_D} = 2\sqrt{(0.8)(0.4)} = 1.13 \text{ mA/V}$$

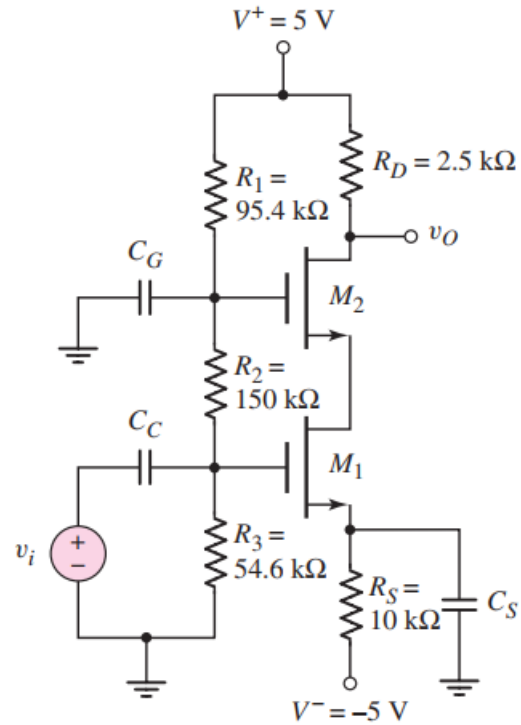
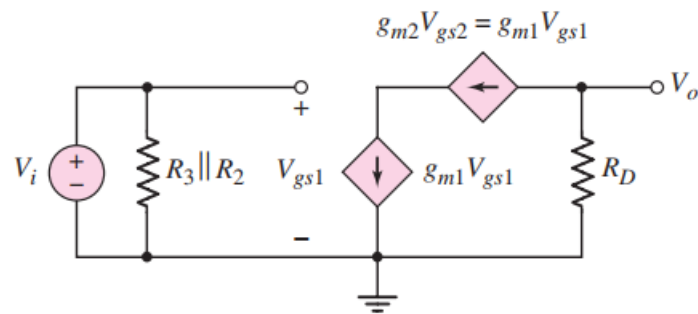


Figure 4.10 NMOS cascode circuit

Small-signal equivalent circuit of NMOS cascode circuit



$$A_v = \frac{V_o}{V_i} = -g_{m1} R_D$$

$$A_v = -(1.13)(2.5) = -2.83$$